NETWORK INTERFACE AND SECURITY ENHANCEMENTS

SECURITY GROUP

1. it is a virtual firewall works at ENI level. (instance level)
2. You can attach multiple Security Groups to one EC2 instance (up to 5 by default).
3. can only have permit rules, cannot have deny rule.
4. It is a stateful, return traffic is automatically allowed in outbound rule, even if there are no rules to allow it.

NACL (network access control list)

1. NACL is an optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets.
2. it works at subnet level.
3. It is a stateless, return traffic is not automatically allowed
4. Supports both allow as well as deny the rule

ELASTIC IP

1. Elastic IP is nothing but a static public IPv4 address provided by amazon web services
2. commonly used for websites, APIs, or SSH access where a stable IP is required.
3. AWS charges for unused Elastic IPs to prevent waste, so you should only allocate when needed.

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PRACTICAL ON ELASTIC IP

1.Create A Instance

1. name ----------> webserver
2. machine --------> ubuntu
3. instance type --> t3. micro
4. create a instance

2.Allocate Elastic Ip Address

1. Go to Elastic IPs section
2. Click Allocate Elastic IP Address
3. Click Allocate

3.Associate Elastic IP with Instance

1. Select the allocated Elastic IP Address
2. Click on Actions → Associate Elastic IP Address
3. Choose:
   - Resource Type → Instance
   - Instance → webserver
4. Click Associate

4.Verify

1. Now your instance has a static public IP Address
2. Stop and start the instance → Elastic IP will NOT change
3. (Normal public IP changes, but Elastic IP stays same)

PLACEMENT GROUPS

A placement group is a concept used in cloud computing to control how your compute resources (like virtual machines/instances) are physically placed on underlying hardware.

1. Low latency (fast communication between instances)
2. High availability (avoid failures affecting multiple instances)
3. High throughput (better network performance)

Types of placement groups

1. **Cluster Placement Group**
   1. Places instances very close together in the same Availability Zone.
   2. Achieve low latency and high network throughput.
   3. Uses
      1. High-performance computing (HPC)
      2. Big data workloads
      3. Distributed databases
   4. It is Best when you need fast communication between instances.

2. **Spread Placement Group**
   1. Places instances far apart across different hardware (racks).
   2. Reduce risk of simultaneous hardware failure.
   3. Max 7 instances per AZ.
   4. Use cases:
      1. Critical applications
      2. Small number of important instances
   5. Best for high availability.

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3. **Partition Placement Group**
    1. Divides instances into logical partitions, each on separate hardware.
    2. Limit failure impact to only one partition.
    3. Use cases:
        1. Large distributed systems like Hadoop, Kafka, Cassandra
    4. Balance between performance and fault tolerance.

WHAT IS NIC CARD

1. NIC is a network interface card
2. It is hardware device that allows a computer to connect to a network or the internet.
3. Without a NIC, a computer cannot communicate with other devices on a network.

What NIC DO

- Provides a physical connection to the network
- Has a MAC address (unique identity of the device)
- Converts data into signals for transmission

Types of NIC

1. internal network card
2. External network card

Internal Network Card

1. Installed inside the computer (on motherboard or PCI slot)
2. Mostly used in desktops
3. Uses Ethernet cable (RJ45) for connection
4. Today, many motherboards have built-in NIC (LAN port)



External Network Card

1. Used when internal NIC is not available or not working
2. Connected via USB port

3. Common in laptops or compact devices
4. It is two types:
    - **Wired (USB to Ethernet)** → uses LAN cable
    - **Wireless (Wi-Fi adapter)** → connects without cable



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